NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2024-2025

IE2104 – DIGITAL ELECTRONICS

April / May 2025 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Given two negative numbers, -337_{10} and -106_8,

   (i) express the two numbers in 2's complement 10-bit binary;

   (ii) perform the addition of the two negative numbers. Show your workings and verify your answer in decimal.

   (8 Marks)

   (b) Given $f(a,b,c,d) = (a+c+d)(\overline{b}+c+\overline{d})(a+\overline{c}+\overline{d})$,

   (i) express the function in sum of minterms format, i.e., $f(a,b,c,d) = \sum m(\ldots)$;

   (ii) find the minimized sum of products expression of $f(a,b,c,d)$;

   (iii) draw the minimum 2-level logic circuit using only NAND gates. You may assume that the complemented inputs are available, i.e., $\overline{a}$, $\overline{b}$, $\overline{c}$, and $\overline{d}$.

   (9 Marks)

   (c) Without modifying the logic function and with minimum number of transistors, draw a CMOS circuit for $Y(a,b,c,) = \overline{(a \cdot c)} \cdot (a+b+c)$.

   (8 Marks)

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2. (a) The function table of a 3-to-8 decoder is given in Table 1. Draw a truth table for a 1-bit Full Adder and design the circuit using ONLY ONE 3-to-8 decoder and minimum number of logic gates. Show your circuit diagram.

(8 Marks)

Table 1

Inputs			Outputs							
A	***B***	***C***	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

(b) The function table of a 2-to-1 multiplexer is given in Table 2. Using TWO of these multiplexers and minimum number of logic gates, re-construct the 1-bit Full Adder circuit. Show your circuit diagram.

(10 Marks)

Table 2

S	***Y***
0	D_0
1	D_1

Note: Question No. 2 continues on page 3.

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(c) Using the Programmable Logic Array (PLA) circuit of Figure 1 and ONLY ONE 2-input XOR gate, implement the function $F(a,b,c,d) = \prod M(4,5,6,7,9,10,12,15)$. Show your circuit diagram with clearly labelled inputs and output.

(7 Marks)

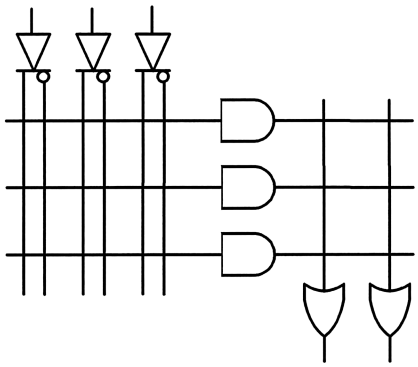


Figure 1

3. A vending machine has two inputs, X and Y, and two outputs, R and Z. The vending machine accepts only two types of coins: 50-cent and $1. Only one coin can be inserted at any one time, and no more than $2 can be deposited on a single purchase. If no coin is inserted, XY = "00". If a 50-cent coin is inserted, XY = "01". If a $1 coin is inserted, XY = "10". If the reject button is pressed, XY = "11". When the reject button is pressed, the transaction is cancelled, and the machine asserts the output R to return all coins that have been deposited in the current transaction. When exactly $1.50 is deposited, the machine asserts the output Z to dispense a can of drink. If more than $1.50 is deposited, the machine asserts both R and Z at the same time to dispense the drink and return the change. The asserted outputs are automatically reset to 0 in the next clock cycle.

(a) Figure 2, on page 4, shows an incomplete state diagram of the vending machine. Complete the state diagram by adding in the missing state transitions.

(5 Marks)

(b) Identify the type of machine (Moore or Mealy) that the state diagram is designed for and state your reason.

(4 Marks)

Note: Question No. 3 continues on page 4.

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(c) Design the machine using D flip-flops and basic gates with any number of inputs. Use A and B as the state variables and the following state assignments:

S_0: AB = "00"
S_1: AB = "01"
S_2: AB = "10"

(i) Tabulate the characteristic and excitation table.
(ii) Use K-maps to obtain the minimized logic functions of the next state and the output decoders.
(iii) Draw the circuit diagram.
(iv) Determine the next state AB and outputs RZ of the unused state of the circuit when the inputs XY = "01".

(16 Marks)

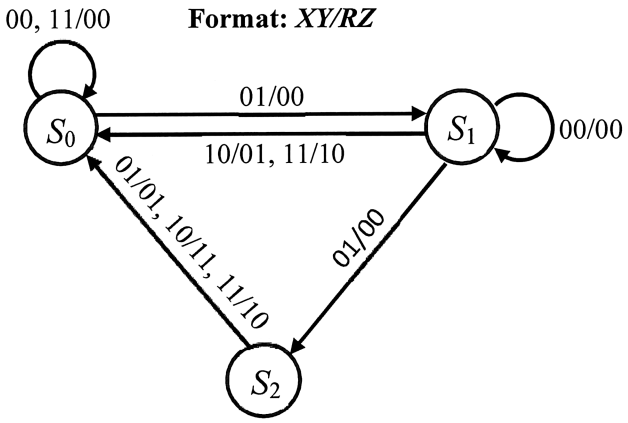


Figure 2

4. (a) A clock generator circuit and the delay of each of its gate are shown in Figure 3, on page 5. The input W is initially at logic '0' and it switches to logic '1' after 5 ns. The timing waveform of W from 0 ns to 100 ns is also shown in Figure 3.

(i) Sketch the waveforms of X, Y and Z from 0 ns to 100 ns?
(ii) Determine the frequency and duty cycle of the output signal Z.
(iii) What is the function of the input W?

(10 Marks)

Note: Question No. 4 continues on page 5.

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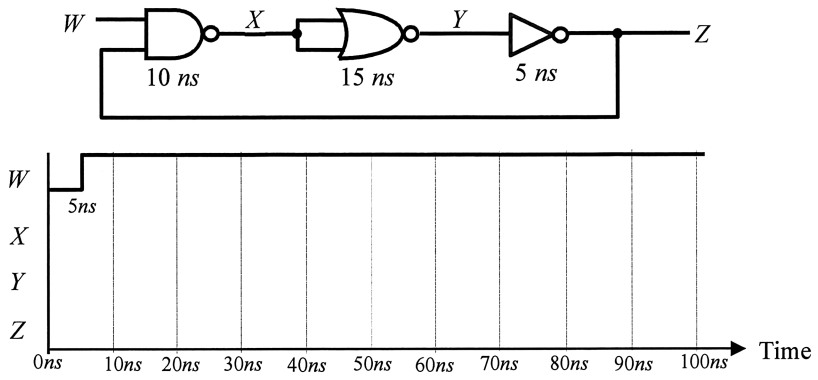


Figure 3

(b) Figure 4 shows the block diagram of a DRAM chip.

(i) What are the functions of RAS_L, CAS_L, and WE_L?

(ii) Draw the circuit of a single bitcell.

(iii) What is the size of this memory chip?

(iv) What is the number of rows and the number of columns of the bitcell array?

(v) How many refresh cycles are required for the whole chip?

(vi) If each memory cell needs to be refreshed every 50 ms, what is the refresh period for each cell?

(vii) If the refresh time is 40 ns, determine the percentage of time the chip expended on refreshing?

(15 Marks)

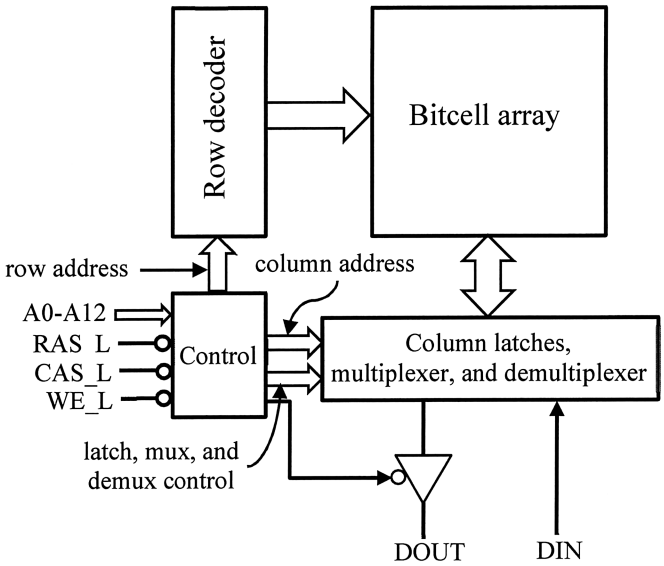


Figure 4

END OF PAPER

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2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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